

Program 4: Integrate Git with Docker for Source-Controlled Application Builds

Project Structure:

Program -4/

→ Dockerfile

→ [index.js](#)

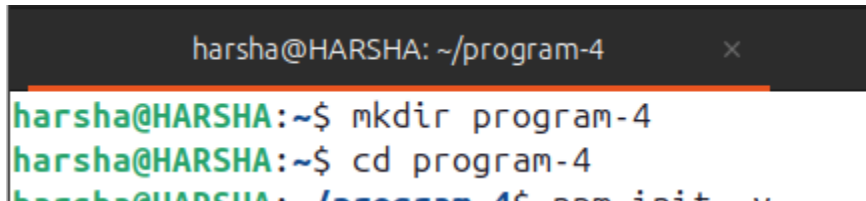
→ package.json

→ package-lock.json

Step 1: Create Project Folder

mkdir program-4

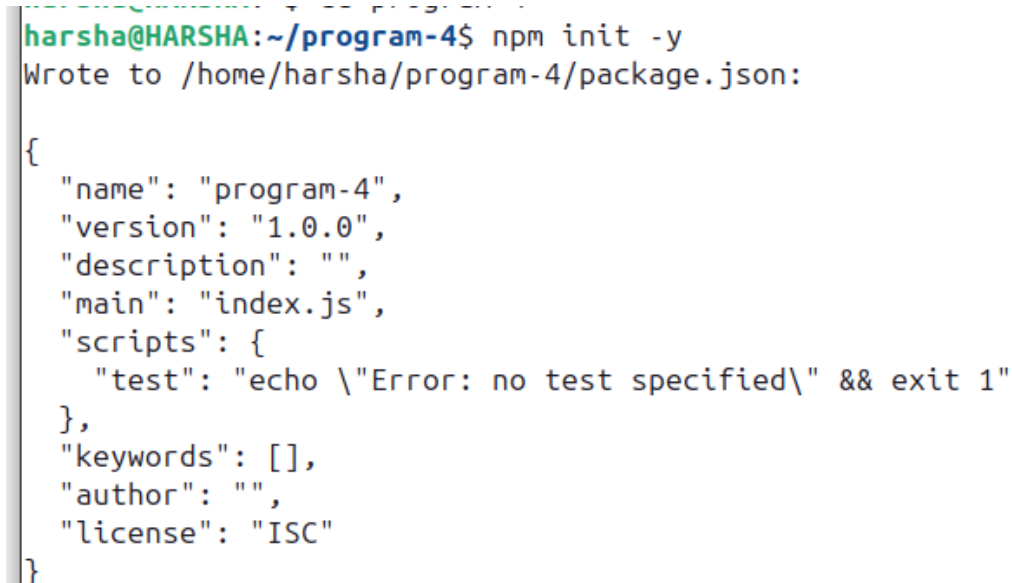
cd program-4

A terminal window titled 'harsha@HARSHA: ~/program-4' with a close button. The terminal shows the commands 'mkdir program-4' and 'cd program-4' being executed successfully. The prompt changes from '~\$' to '~/program-4\$' after the second command.

```
harsha@HARSHA: ~/program-4
harsha@HARSHA:~$ mkdir program-4
harsha@HARSHA:~$ cd program-4
harsha@HARSHA:~/program-4$
```

Step 2: Initialize Node Project

npm init -y

A terminal window showing the execution of 'npm init -y'. It displays the message 'Wrote to /home/harsha/program-4/package.json:' followed by the generated package.json content in JSON format.

```
harsha@HARSHA:~/program-4$ npm init -y
Wrote to /home/harsha/program-4/package.json:

{
  "name": "program-4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Step 3: Create index.js

nano [index.js](#)

```
GNU nano 7.2 index.js
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req, res) => {
  res.send("Hello from Git + Docker Integration!");
});

app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

Step 4: add the dependency

npm install express

Step 5: Create Dockerfile

nano Dockerfile

```
GNU nano 7.2 Dockerfile
# Stage 1 - Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

# Stage 2 - Runtime Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]
```

Step 6: Create a GitHub Repository

Open GitHub → Click **New Repository**

Give a repository name (e.g., **program-4**)

STEP 7: Get a Personal Access Token (PAT)

Profile → Developer Settings → Personal Access Tokens → Tokens (classic)

STEP 8: Initialize Git in Project

git init

git add .

git commit -m "Initial commit"

git branch -M main

```
harsha@HARSHA:~/program-4$ git init
git add .
git commit -m "Initial commit"
git branch -M main
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/harsha/program-4/.git/
[master (root-commit) 0cbe9e0] Initial commit
3 files changed, 41 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 package.json
```

STEP 9: Add Remote Repo

git remote add origin https://github.com/<username>/<repo>.git

STEP 10: Push Code to GitHub

git push -u origin main

```
harsha@HARSHA:~/program-4$ git push -u origin main
Username for 'https://github.com': harshamuniraju
Password for 'https://harshamuniraju@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 803 bytes | 803.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/harshamuniraju/program-4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

STEP 11: Run Docker Container Locally

docker build -t git-docker-app .

```
harsha@HARSHA:~/program-4$ docker build -t git-docker-app .
[+] Building 14.9s (14/14) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 270B                                0.0s
=> [internal] load metadata for docker.io/library/node:18-slim    12.9s
=> [internal] load metadata for docker.io/library/node:18        12.9s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                   0.0s
=> => transferring context: 32.11kB                                  0.0s
=> [build 1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67 0.0s
=> [stage-1 1/3] FROM docker.io/library/node:18-slim@sha256:f9ab18e354e6 0.0s
```

docker run -p 3000:3000 git-docker-app

```
=> => naming to docker.io/library/git-docker-app                  0.0s
harsha@HARSHA:~/program-4$ docker run -p 3000:3000 git-docker-app
Server running at http://localhost:3000
```

Out Put

Browser → <http://localhost:3000>

